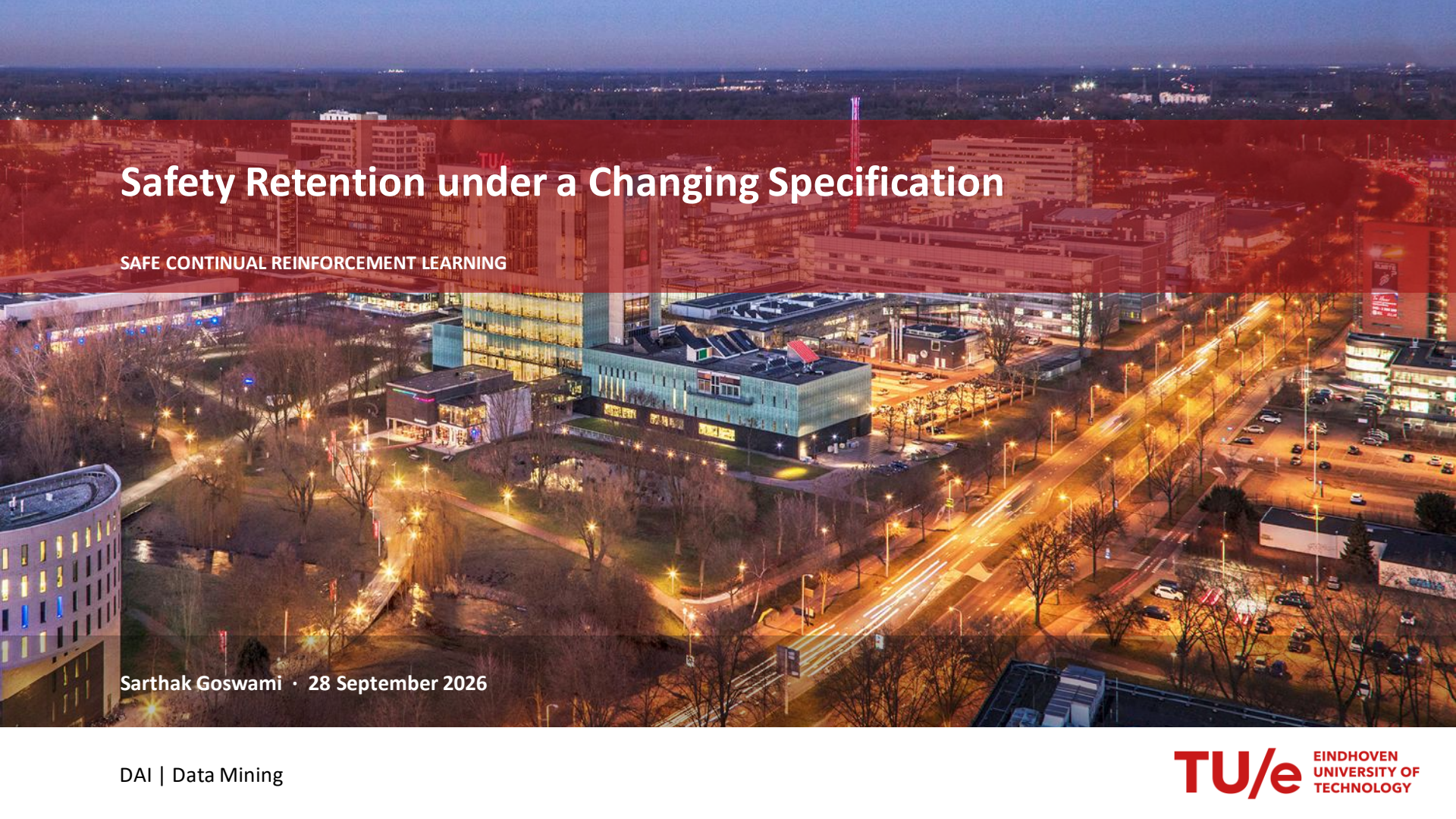


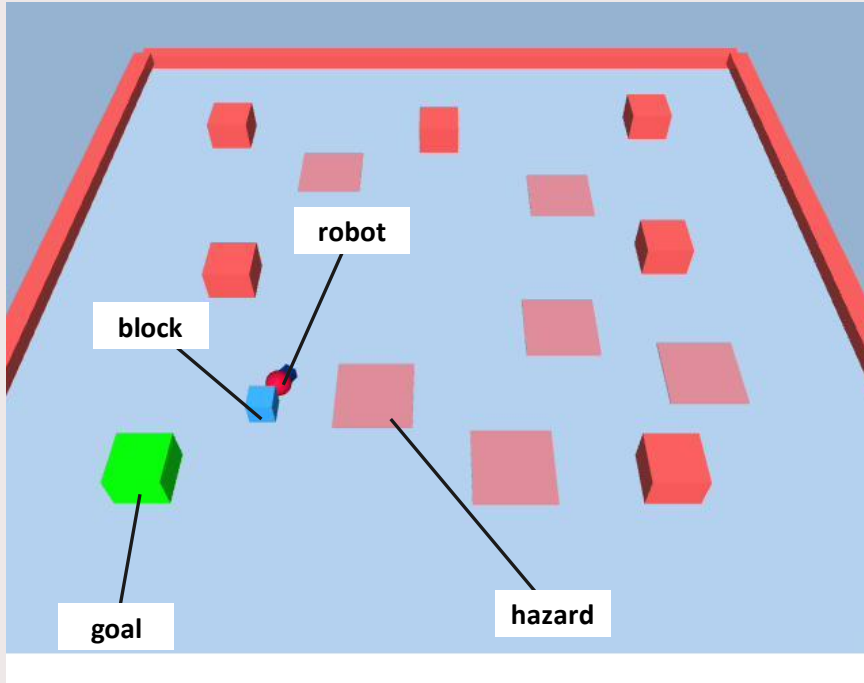


Safety Retention under a Changing Specification

SAFE CONTINUAL REINFORCEMENT LEARNING

Sarthak Goswami · 28 September 2026

A warehouse robot's safety limit changes



In this thesis: a simulated robot pushes a block to a goal (CRAX Push, Tomilin et al., 2026)

Trial
lenient

≤ 25 collisions
per shift

↓ retrain

After an incident
tightened

≤ 5 collisions
per shift

↓ retrain

Proven safe again
loosened back

≤ 25 collisions
per shift

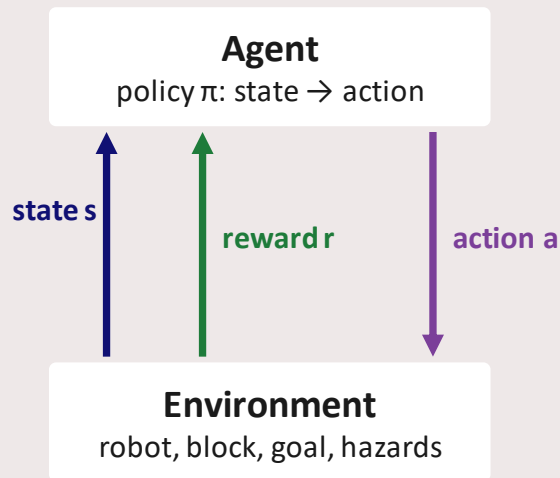
After each retraining, does the robot still meet the **earlier** limits?

A certificate covers the controller that was tested (Hawkins et al., 2021).

Part 1 · Background

Reinforcement learning, safe RL, continual RL, and the gap between them

RL: learn a policy that maximises reward



$$M = (\mathbf{S}, \mathbf{A}, \mathbf{P}, \mathbf{r}, \gamma, \rho_0)$$

S **states**: sensor readings

A **actions**: motor commands

P **dynamics**: how the world responds

r **reward**: progress to the goal

γ **discount**: later reward counts less

ρ_0 **start**: where episodes begin

$$J_r(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_t \gamma^t r(s_t, a_t) \right]$$

$$\pi^* = \arg \max_{\pi} J_r(\pi)$$

The policy maximises expected discounted reward.

Safe RL: maximise reward, keep cost under d

$$M = (S, A, P, r, \mathbf{c}, \gamma, \rho_0, \mathbf{d})$$

c **cost:** contact with a hazard, the story's collisions

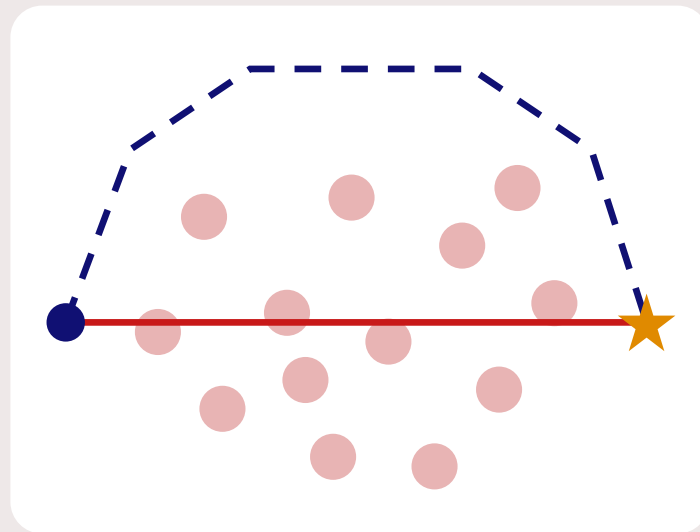
new

d **bound:** the most average cost allowed per episode

new

$$\max_{\pi} J_r(\pi) \quad \text{subject to} \quad J_c(\pi) \leq d$$

- **Certified:** on test episodes, average cost $\leq d$
- **P3O** (method used): a penalty with weight κ that rises while cost exceeds d



--- $d = 5$: long detour, low cost

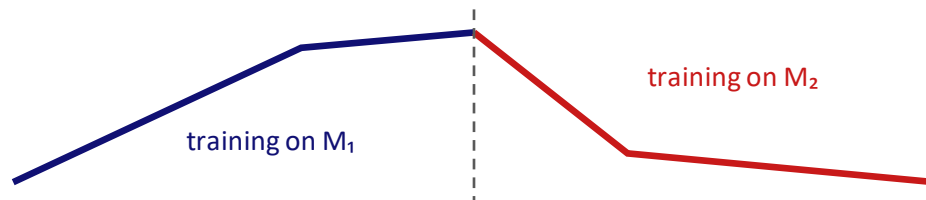
— $d = 25$: shorter route, more reward

Continual RL: new training erodes old skills



Tasks arrive one after another; earlier tasks are not trained again

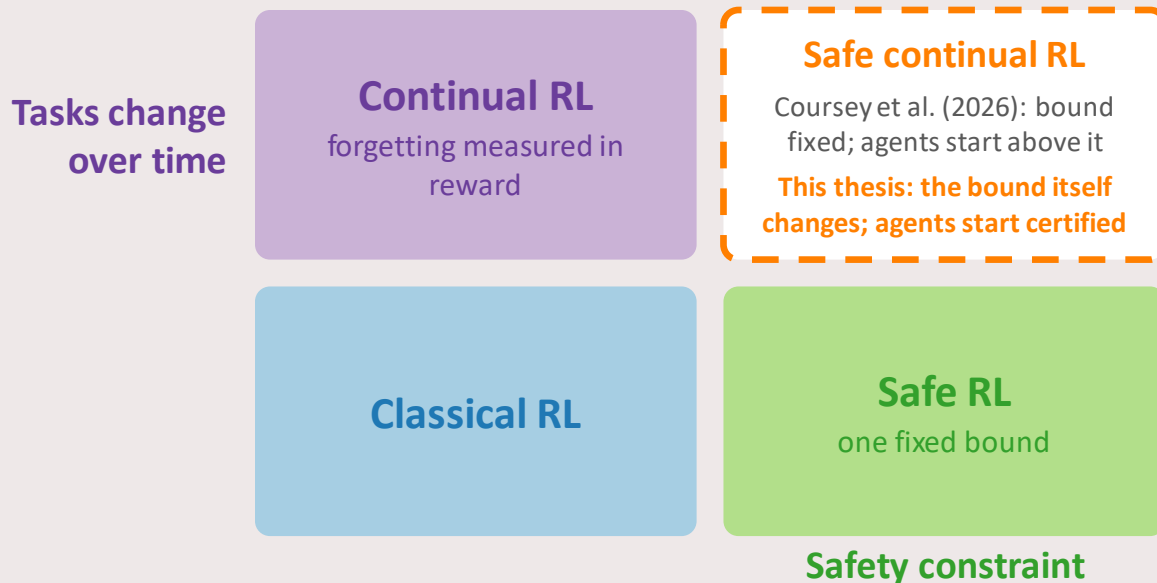
Reward on M_1 (schematic): how forgetting is usually measured



Remedies: regularisation (EWC, used here), replay, parameter isolation.

Catastrophic forgetting: training on M_2 lowers performance on M_1 .

The gap: a bound that changes over time



No prior work tests whether a certified bound survives retraining under another bound.

Coursey et al. (2026) keep the bound fixed

	Coursey et al. (2026)	This thesis
Environment changes over time	✓ damaged robots, a mug-carrying arm	✓ bound, level, or task
The bound itself changes	✗ fixed throughout	✓ $5 \leftrightarrow 25$, and three-phase sequences
Safe RL method inside the retraining	✗ no safe RL algorithm: PPO + EWC, not a constrained method	✓ P3O with EWC
Agents safe before the change	✗ costs far above their bounds	✓ certified: 0.86–0.99 of d
Safety checked against the old bound	✗ total cost during training, not vs a bound	✓ test cost ÷ old bound

Open question: does a certified bound survive retraining under a new one?

Research question

When the setting a safe agent was certified in changes, what does it forget, what does it retain, and does the answer depend on what kind of change occurred?

SQ1 · Threshold

the bound changes

SQ2 · Difficulty

the environment changes

SQ3 · Task

the objective changes

Part 2 · Method and testbed

The setting, the benchmark and its levels, the protocol, and EWC

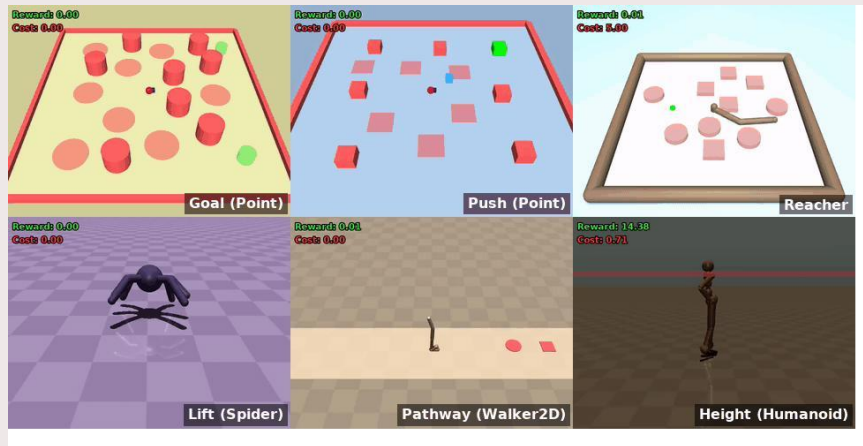
The setting: a sequence of constrained MDPs



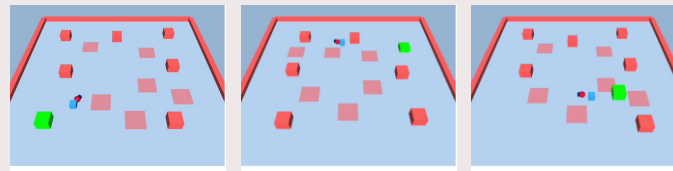
$$M_k = (S, A, \mathbf{P}_k, \mathbf{r}_k, c, \gamma, \rho_0, \mathbf{d}_k)$$

Axis	Changes	Agent told?	Run here
Threshold	bound $\mathbf{d}_k$	Yes: the bound is an input	$d = 5 \leftrightarrow 25$
Difficulty	dynamics $\mathbf{P}_k$	No	5 level pairs, $d = 25$
Task	reward $\mathbf{r}_k$	Partly	Goal $\leftrightarrow$ Push, $d = 25$

CRAX Push: move a block past hazards



CRAX suites (Tomilin et al., 2026). This thesis uses Goal and Push.



Push L1
goal speed 0.0

Push L2
goal speed 0.3

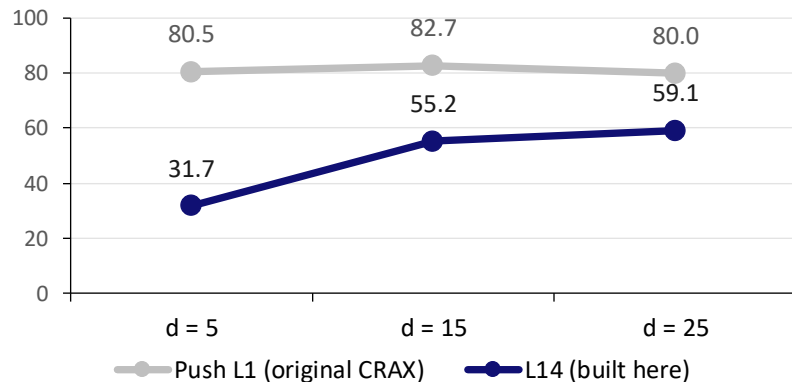
Push L3
goal speed 0.6

- Safe RL benchmark in JAX
- Push: reward for moving the block to the goal, cost for touching a hazard
- Push L1–L3 ship with CRAX; L4–L16 were built here

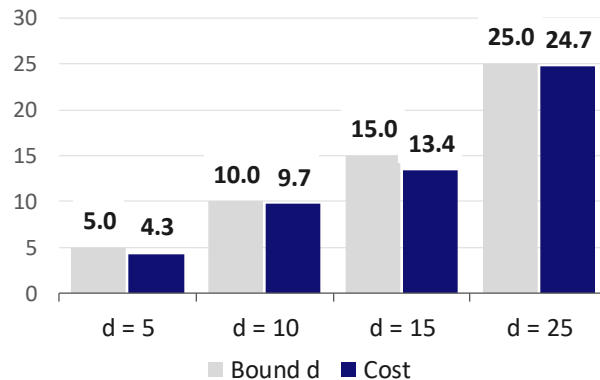
A looser bound must buy reward: L14

If the bound does not change behaviour, a change of bound has nothing to make the agent forget

Reward of agents trained at each bound

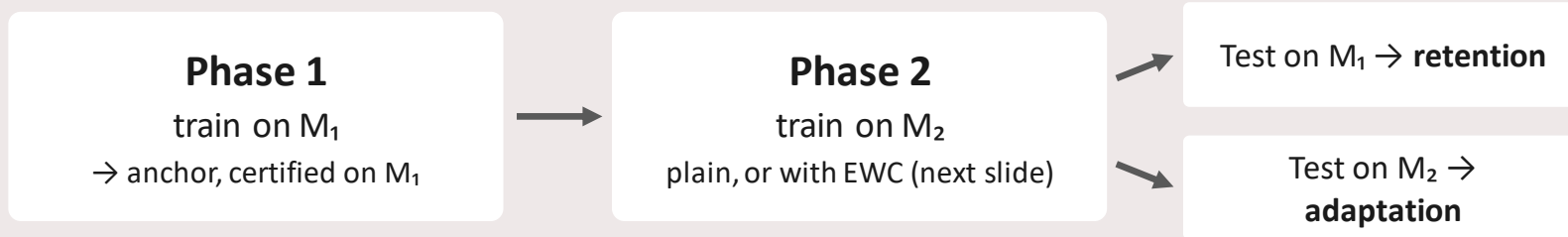


L14: cost of agents trained at each bound



On L14 a looser bound buys 27 reward, and every agent stays under its bound.

Protocol: train, retrain, test on both



$$\rho_{\text{ret}} = \frac{C(\mathcal{M}_1)}{d_1} \quad \rho_{\text{adp}} = \frac{C(\mathcal{M}_2)}{d_2}$$

cost ÷ bound, on the old and on the new task

■ ≤ 1.00 the certificate holds

■ **1.00–1.05** at the bound (the tolerance used to pick λ)

■ **> 1.05** broken

Plain retraining and EWC start from the same anchor · 500M steps per phase · 5 runs.

EWC: a spring ties each weight to the anchor

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$$

P3O loss on the new task (points to $\mathcal{L}_{\text{task}}(\theta)$)
 distance from the anchor's weight (points to $(\theta_i - \theta_i^*)^2$)
 overall strength λ : chosen (0 = plain) (points to $\frac{\lambda}{2}$)
 stiffness of weight i : computed (points to F_i)

Stiffness of weight $i = \lambda \times F_i$



λ : one number for all weights, chosen per transition (appendix A5).

F_i (Fisher information): how much the anchor's action choices change when θ_i moves. Computed from the anchor, not tuned.

θ : the network's weights; θ^* : the anchor's weights. Kirkpatrick et al. (2017).

Part 3 · Results

Threshold · difficulty · task · three phases · across all axes

Axis 1: the bound changes between 5 and 25

Loosening

phase 1
d = 5



phase 2
d = 25



test on both
d = 5 and 25

Tightening

phase 1
d = 25



phase 2
d = 5

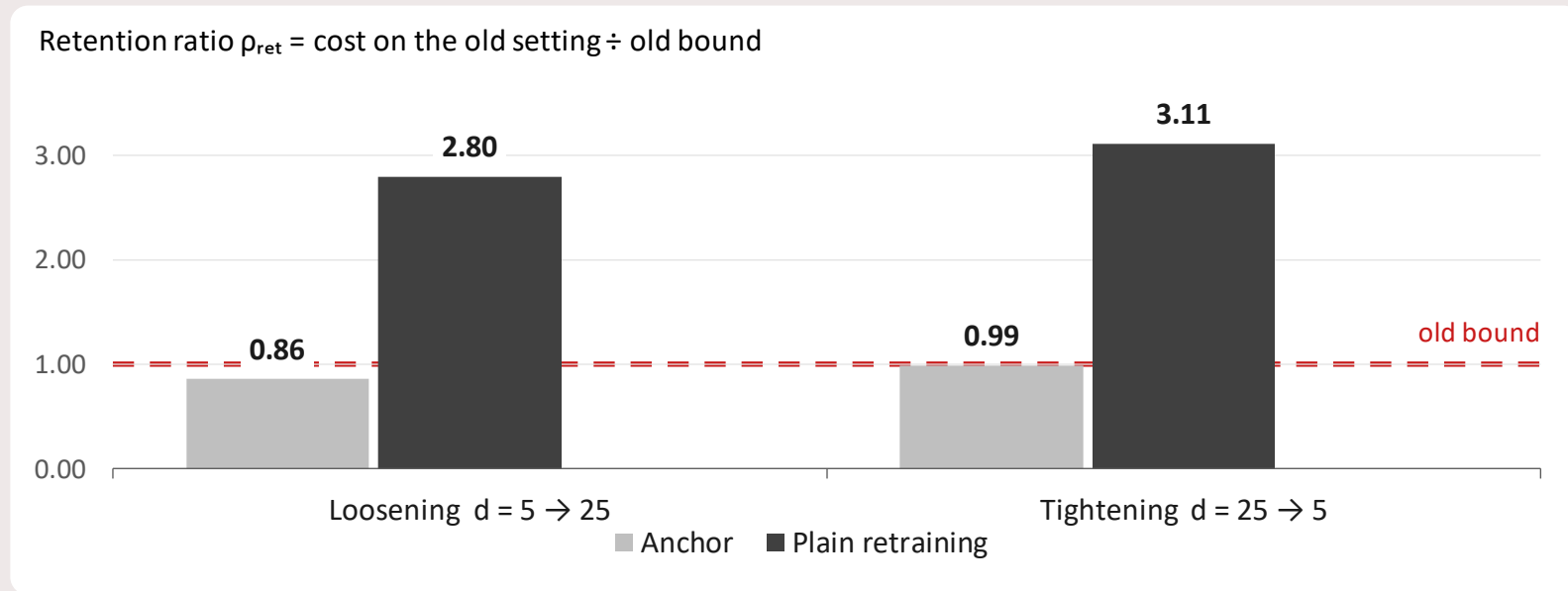


test on both
d = 25 and 5

Level L14 · 5 runs · the bound is an input, so the agent is told the new one

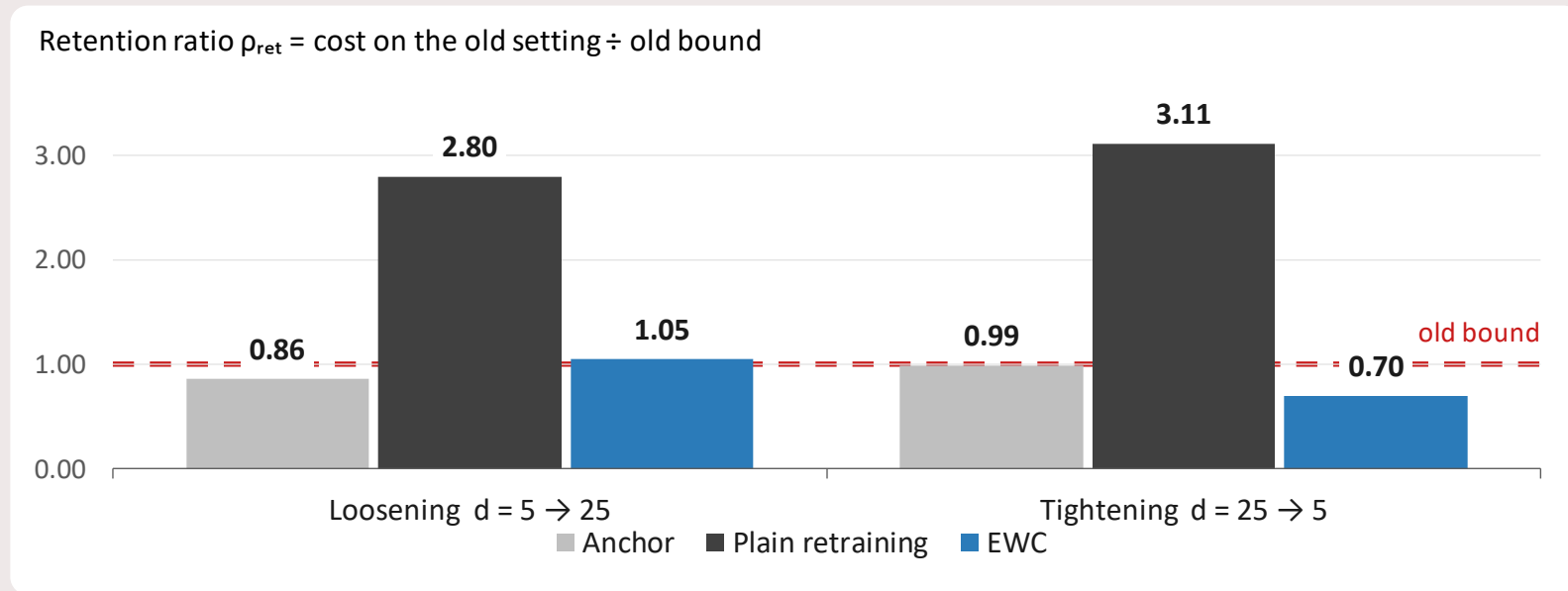
SQ1: told the new bound, does the agent still meet the old one?

Plain retraining breaks the old bound both ways



The old bound is in the agent's input throughout, and it still breaks it about threefold.

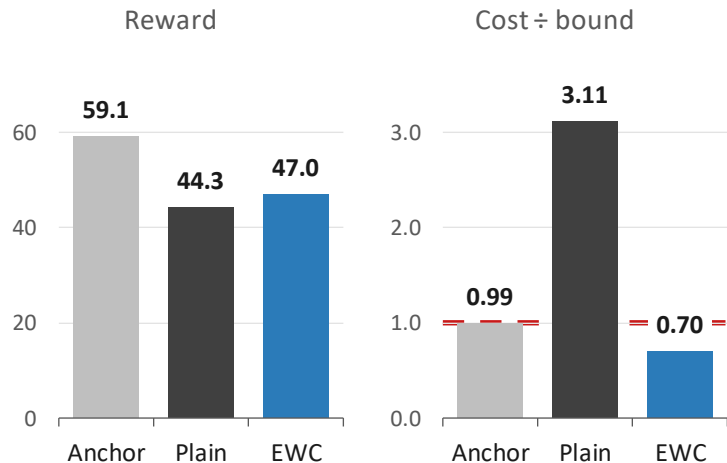
EWC: 0.70 × when tightening, 1.05 × when loosening



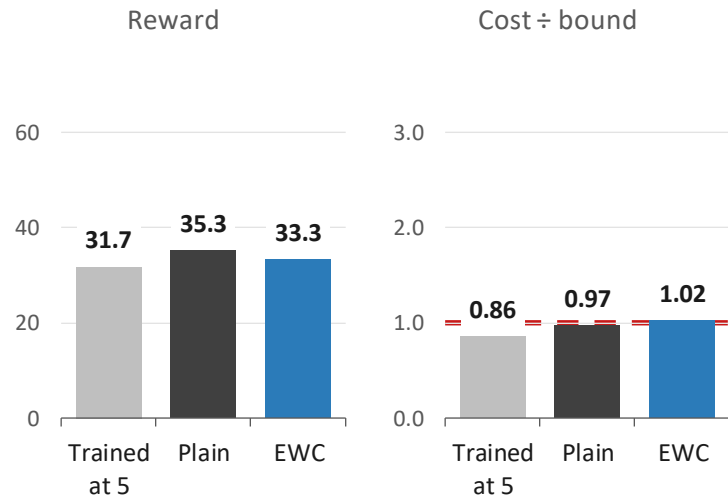
Tightening: well under the old bound. Loosening: at it, within the run-to-run spread (5.24 ± 1.74 vs 5).

Tightening: EWC keeps the old bound and adapts

At the old bound, $d = 25$ (retention)



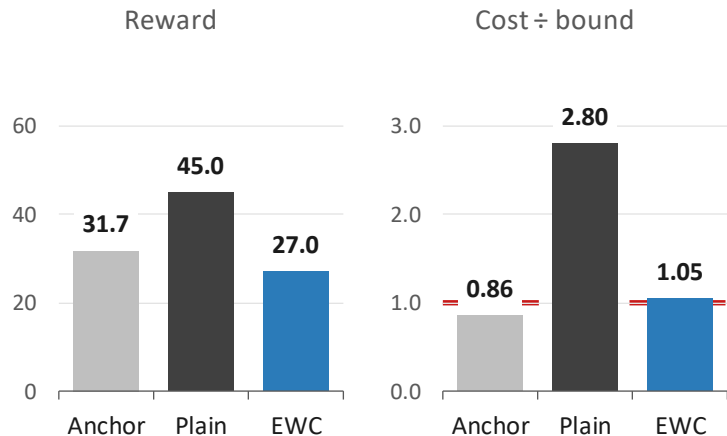
At the new bound, $d = 5$ (adaptation)



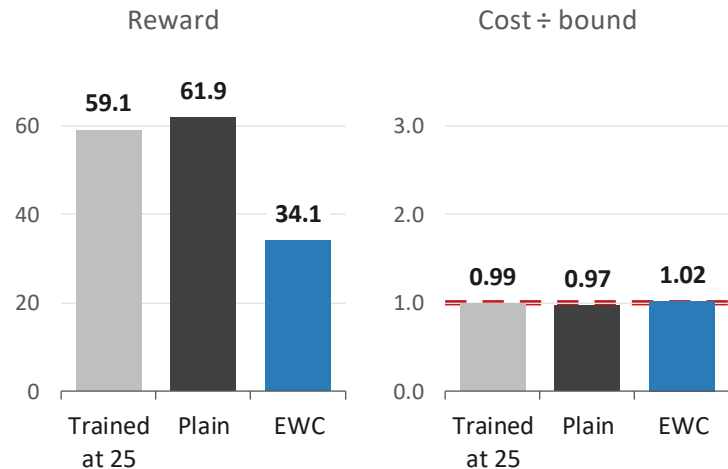
EWC meets both bounds, keeps 79% of the old-bound reward, and adapts as well as plain retraining.

Loosening: at the old bound, but at a price

At the old bound, $d = 5$ (retention)



At the new bound, $d = 25$ (adaptation)



EWC: 5.24 ± 1.74 vs 5, inside the run-to-run spread. Plain earns more old-setting reward (45.0 vs 31.7) via riskier routes.

SQ1: EWC keeps the old bound both ways; after loosening only at its edge, with 45% less new reward.

Why the directions differ: which way EWC pulls

Tightening 25 → 5



P3O penalty: strong

Reward

EWC: toward the anchor

EWC and reward agree: keeping the old bound is cheap

Loosening 5 → 25



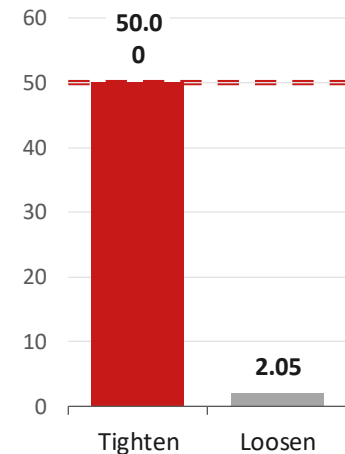
P3O penalty: weak

Reward

EWC: toward the anchor

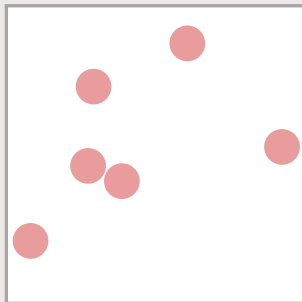
Only EWC holds back reward: keeping the old bound costs reward

Median κ , 163 runs



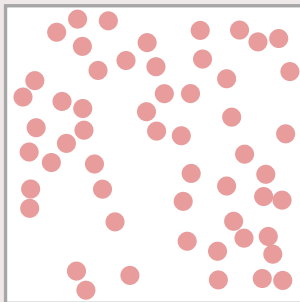
κ grows only while cost exceeds the bound, so it stays small when loosening. Alternative: the cap at 50.

Axis 2: five levels, bound fixed at 25



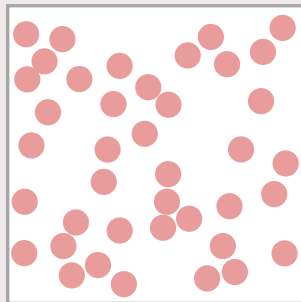
L1 original CRAX

6 hazards · goal speed 0



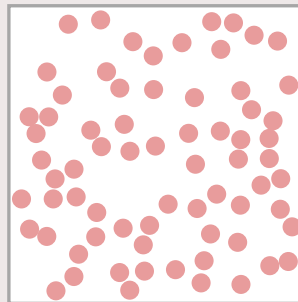
L8 many small

55 hazards · goal speed 0.50



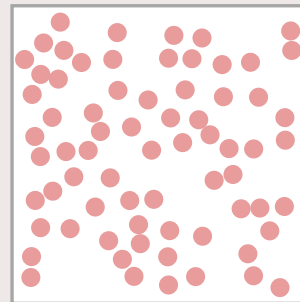
L11 fewer, larger

40 hazards · goal speed 0.50



L14 densest

70 hazards · goal speed 0.50



L16 faster goal

70 hazards · goal speed 0.65

SQ2: the agent is not told the level changed. How much safety is lost, and can it be predicted?

Testing existing agents predicts the loss

1 · Test every anchor on every level (cost, bound 25)

tested on →

		L1	L8	L11	L14	L16
trained on ↓	L1	27.2	172.9	219.2	236.5	245.6
	L8	28.5	24.6	65.1	37.4	35.3
	L11	14.9	10.2	23.1	17.9	16.9
	L14	26.0	17.0	55.0	23.6	22.9
	L16	41.9	17.6	57.1	30.6	23.2

Green = new level, navy = old level. Red > 25, blue < 25. Outlined = the transitions at right.

2 · Transitions run, spanning the predicted loss

old level → new level

L14 → L1	9.46	few large, still goal
L11 → L14	2.20	more, smaller hazards
L14 → L8	1.50	fewer hazards
L14 → L16	1.22	faster goal only
L8 → L11	0.41	fewer, larger hazards

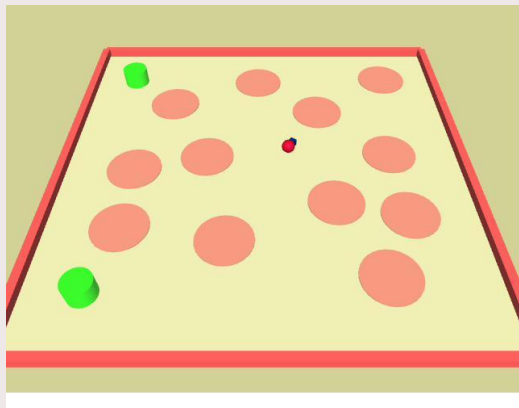
Plain retraining keeps almost none of the old level's safety; the loss can be read off in advance.

EWC restores 3 of 4 breaches and still adapts

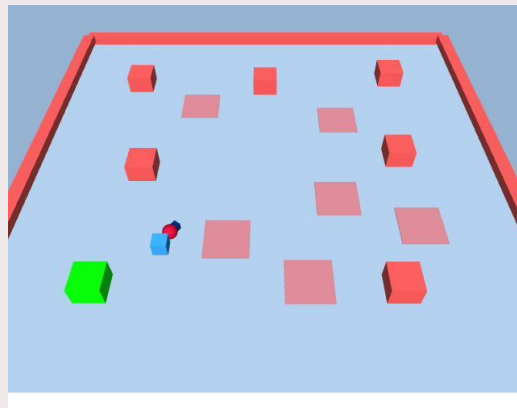
Transition	Predicted cost ÷ bound	Old level (retention)		New level (adaptation)		Old bound
		cost ÷ bound	reward %	cost ÷ bound	reward %	
L14 → L1	9.46	8.61 → 0.82	61% → 82%	0.95 → 0.79	60% → 79%	Restored
L11 → L14	2.20	2.28 → 1.31	112% → 97%	0.92 → 0.95	107% → 86%	Still above
L14 → L8	1.50	1.51 → 0.97	113% → 94%	0.98 → 0.65	110% → 92%	Restored
L14 → L16	1.22	1.07 → 0.95	107% → 94%	1.03 → 0.92	114% → 98%	Restored
L8 → L11	0.41	0.42 → 0.39	93% → 82%	0.98 → 1.00	103% → 92%	Never broken

L11 → L14 stays above: on L11, every other level's anchor costs 55 or more, over twice the bound.

Axis 3: switching between Goal and Push



Goal: reach the goal



Push: push the block

phase 1
Goal

phase 2
Push

test
both

phase 1
Push

phase 2
Goal

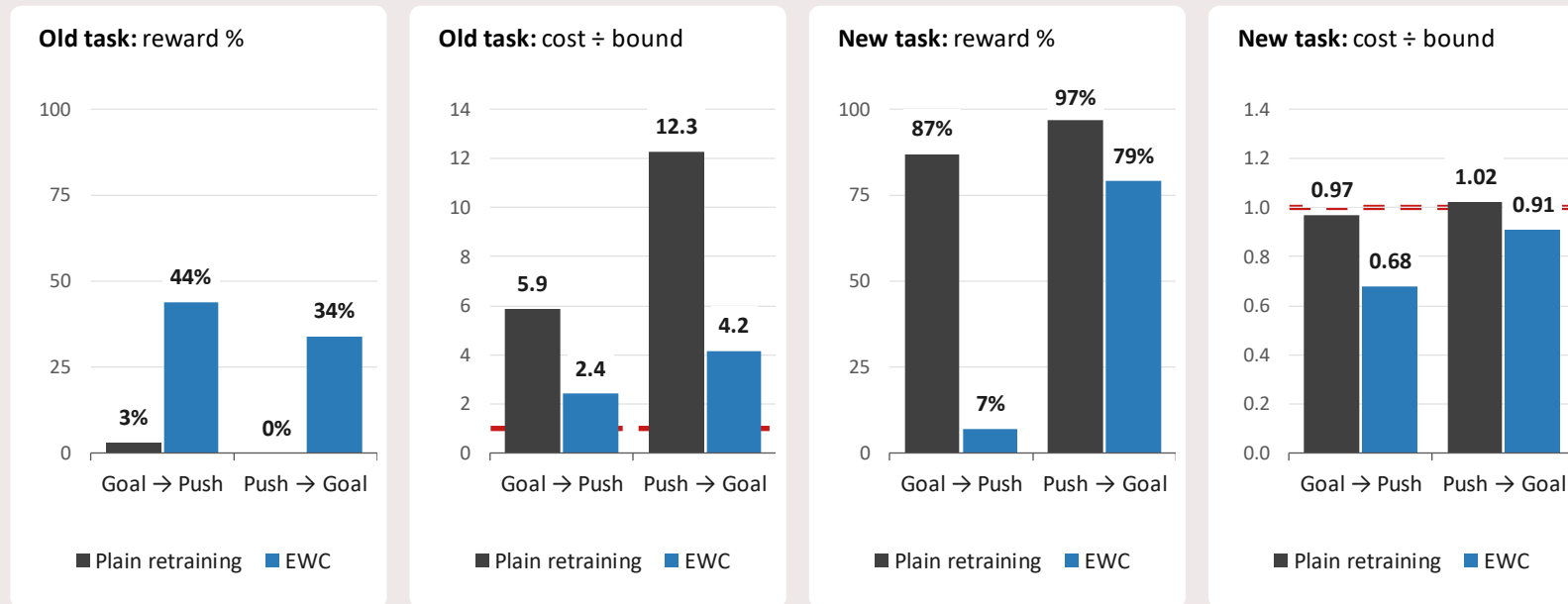
test
both

bound 25 · 3 runs

SQ3: are task skill and safety retained together?

Images: Tomilin et al. (2026). Observations padded so one network serves both tasks.

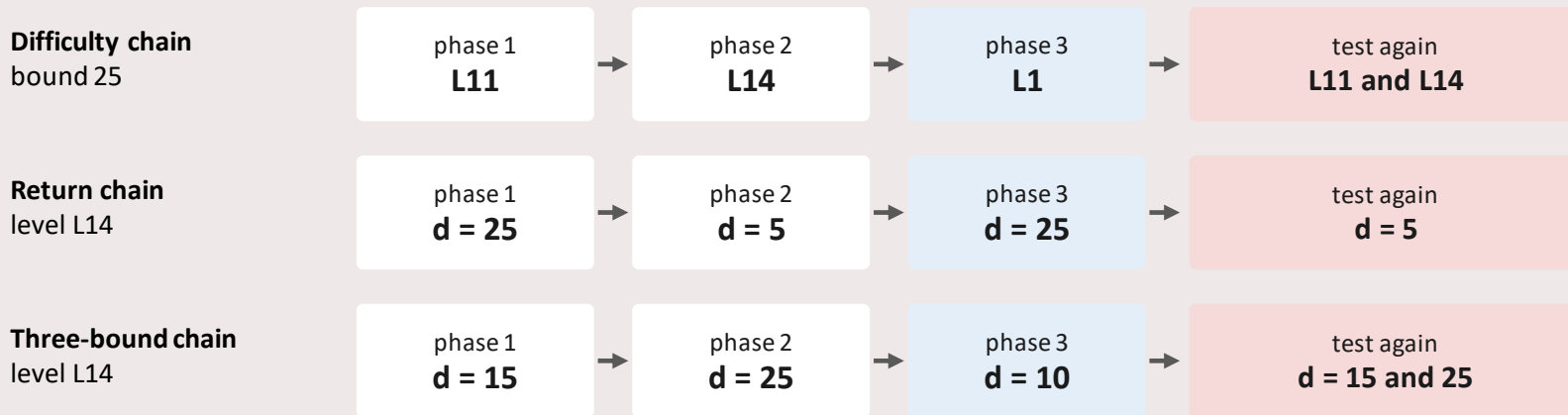
Old-task reward partly returns; cost stays high



SQ3: no. Both arms meet the new task's bound, but old-task cost stays at about 2 × the bound or more.

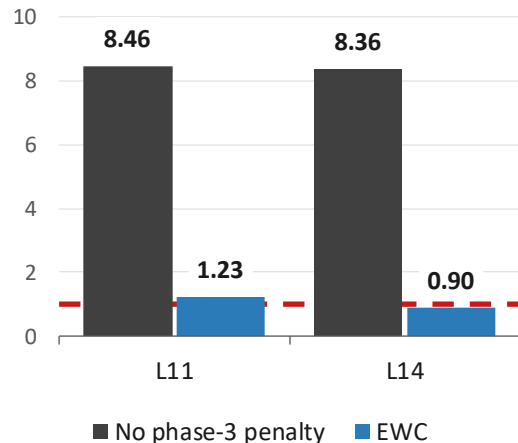
Three phases: three chains, two arms

Does a third phase undo what the first two kept?

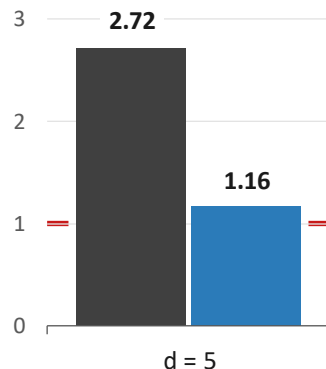


No phase-3 penalty: earlier bounds break again

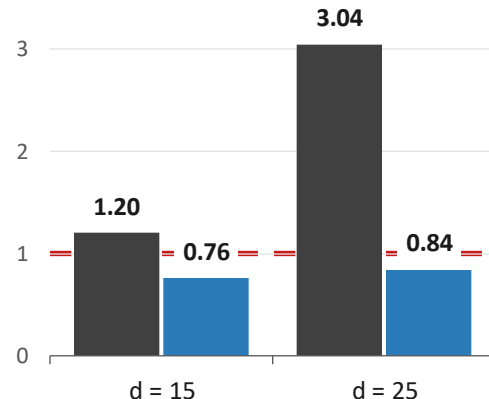
Difficulty chain



Return chain



Three-bound chain



Cost ÷ bound on each earlier setting after phase 3; dashed: bound. Three-bound chain: d = 25 was met after phase 2 (0.99). With EWC, cost on L11 keeps falling during phase 3 (36.57 → 30.22).

Without the penalty: 1.20–8.46 × the bound. With EWC: 0.76–1.23 ×.

Which weights are stiff does not matter

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$$

same penalty, same λ ; only F_i changes

Fisher

from the anchor's
action choices
(standard EWC)

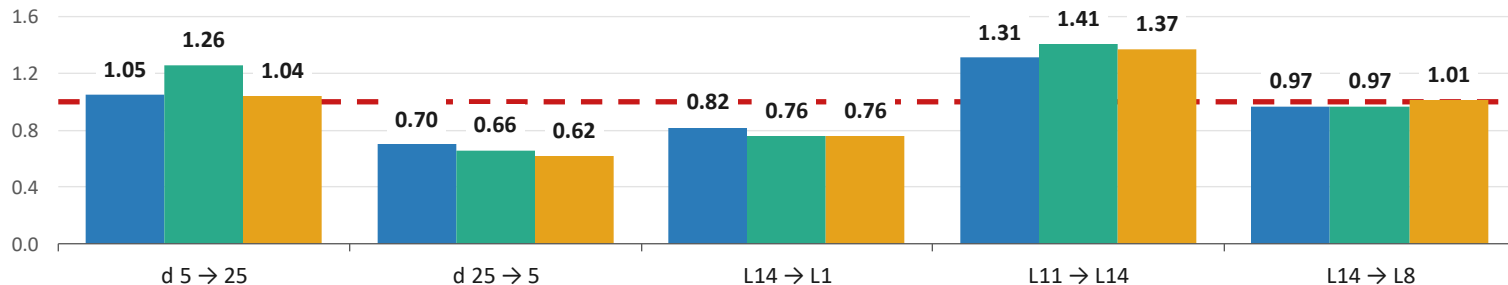
Cost-aware

Fisher weighted by the
cost advantage (not CF-
EWC)

Uniform

$F_i = 1$: every weight
equally stiff

Retention ratio ρ_{ret} per transition



Uniform stiffness retains as well as Fisher: knowing which weights matter does not help here.

On safe agents, CF-EWC is plain EWC

Their finding

Coursey et al. (2026) propose **CF-EWC**: each Fisher sample is weighted by $w = 1/(1+c)$, so costly samples count less.

They ran two arms, **EWC** and **CF-EWC**; CF-EWC matched plain PPO+EWC, with **no explanation**.

The reason

$$w = \frac{1}{1+c}$$

$c = 0$ on 94–99% of samples of a safe agent,

so $w = 1$ almost everywhere,
and the weighted Fisher equals the standard one.

Our measurement

1.0000

Cosine similarity between the two F_i maps, over all **76 anchors**.

It is the angle between the two **5,572-long Fisher vectors**; **1.0** means every weight gets the same relative stiffness.

CF-EWC cannot differ from EWC once the agent is safe, which explains their null result.

Summary of results

	Plain retraining	With EWC	Answer
SQ1 · Threshold	2.80 and 3.11 × the old bound	Tightening: 0.70, both bounds met. Loosening: 1.05, at the bound; 45% less new reward	Being told the new bound does not keep the old one
SQ2 · Difficulty	Up to 8.61 ×, predicted in advance	3 of 4 restored; 79–98% of new-level reward	The loss is predictable
SQ3 · Task	5.86 and 12.27 ×	About 2 × or more; new task can collapse	Reward and safety are forgotten separately
Three phases	1.20–8.46 × on earlier bounds	0.76–1.23 ×	The penalty is needed in every phase
While retraining	1.1–1.9 × (threshold axis)	No lower than plain	EWC protects the result, not the process
Across axes	Reward metrics miss the breach	Choice of F_i makes no difference	Measure retention on cost

Values: cost ÷ bound on the earlier setting. Red: broken (> 1.05) · amber: partly or at the bound · green: kept · grey: not a retention result

Part 4 · Conclusions

Contributions, what it means in practice, limitations and future work

Contributions

1

A setting and an instrument

Constraint non-stationarity, cost-side retention metrics, and the L14 testbed

2

A direction asymmetry

After tightening EWC keeps the old bound almost for free; after loosening only at its edge, with 45% less new reward

3

Reward and safety split

Reward partly returns, cost stays at about $2 \times$ the bound or more; reward metrics miss this

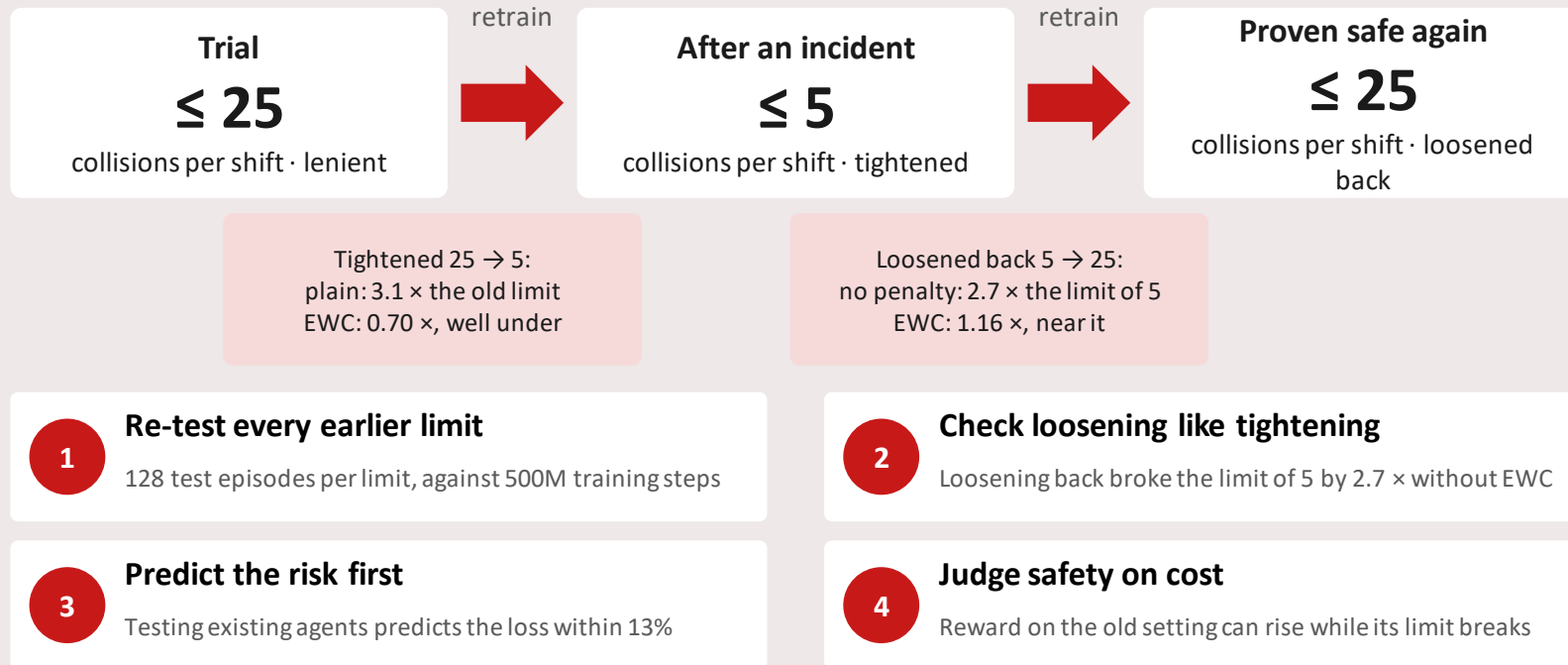
4

A negative result

Uniform stiffness retains as well as Fisher; CF-EWC is plain EWC on safe agents

Also: a predictor of safety forgetting that needs no retraining, and a deployment check that needs only test episodes.

Back to the warehouse robot



Limitations and future work

Limitations

- One algorithm (P3O), one environment family
- Direction result: one pair of bounds
- EWC only, on a small network
- EWC protects the final policy, not cost during retraining

Future work

- Parameter isolation on a larger network
- Size of the bound change
- Longer sequences; a Lagrangian method
- Retention that also limits cost while retraining

Conclusion

The answer depends on the kind of change. The direction of a bound change decides what keeping the old bound costs; task changes retain worst.

SQ1 · Threshold

Plain retraining breaks the old bound. EWC keeps it: freely after tightening, at its edge after loosening.

SQ2 · Difficulty

The loss is predictable; EWC restores three of four broken transitions.

SQ3 · Task

Reward partly returns; cost stays at about $2 \times$ the bound or more.

After retraining, re-test against every bound the agent was certified under, including after a loosening.

Thank you

Questions?

Appendix

Method · testbed · results in full · validity and context

Appendix contents

Method

A1–A7

Testbed

B1–B4

Results in full

C1–C13

Validity and context

V1–V7

A1 · The formal setting

Constrained MDP

$$\max_{\pi} J_r(\pi) \quad \text{subject to} \quad J_c(\pi) \leq d$$

P3O objective

$$\min_{\theta} \mathbb{E}[-A_r^{\pi_k}(s, a)] + \kappa \max\{0, \mathbb{E}[A_c^{\pi_k}(s, a)] + (1 - \gamma)(J_c(\pi_k) - d)\}$$

P3O penalty weight

$$\kappa \leftarrow 1.1 \kappa \quad \text{if violated,} \quad \kappa \leftarrow 0.9 \kappa \quad \text{otherwise,} \quad \kappa \in [0.01, 50]$$

EWC

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$$

Fisher stiffness

$$F_i = \frac{1}{N} \sum_{n=1}^N w_n \left(\frac{\partial}{\partial \theta_i} \log \pi_{\theta^*}(a_n | s_n) \right)^2$$

Three phases

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \frac{\lambda_1}{2} \sum_i F_i^{(1)} (\theta_i - \theta_{1,i}^*)^2 + \frac{\lambda_2}{2} \sum_i F_i^{(2)} (\theta_i - \theta_{2,i}^*)^2$$

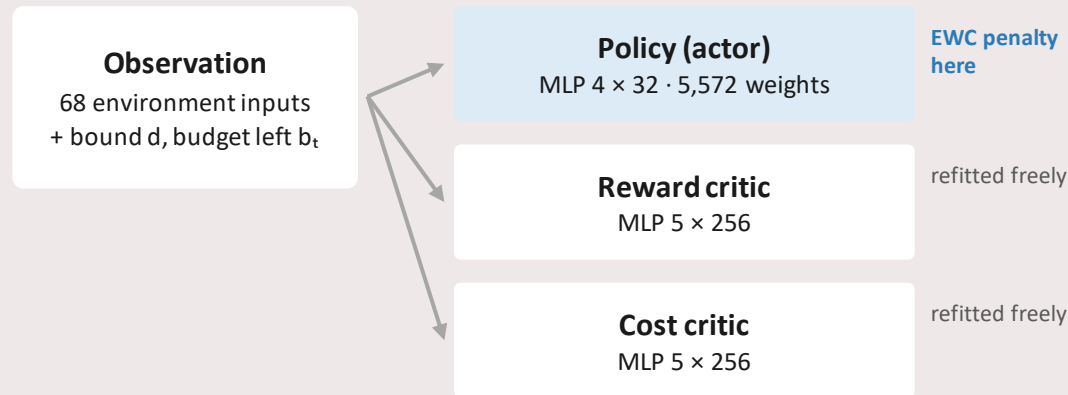
Policy input

$$\pi_{\theta}(a | s, d, b_t), \quad b_t = \text{budget left in the episode}$$

Retention and adaptation

$$\rho_{\text{ret}} = \frac{C(\mathcal{M}_1)}{d_1} \quad \rho_{\text{adp}} = \frac{C(\mathcal{M}_2)}{d_2}$$

A2 · Network and where EWC acts



- Fisher estimated once at the anchor, actions sampled from the anchor policy (the true Fisher), then divided by its mean: every map has mean 1
- Stored between phases: θ^* and the Fisher diagonal
- Observation normaliser restored with the anchor and keeps updating in phase 2
- Phase 3: two terms, λ_1 toward anchor 1 and λ_2 toward the phase-2 output

- Critics are refitted anyway: the values change with the policy on every axis
- Policy-only regularisation is the continual RL convention (MEAL reports critic regularisation can also help)

A3 · What the agent observes

Axis	Task boundary known to the algorithm?	What changes in the observation	Can the policy tell?
Threshold	Yes: EWC starts at the boundary	The bound channel d , and the budget channel b_t	Yes
Difficulty	Yes	Nothing labels the level; only the sensed hazards and goal motion	Only indirectly
Task	Yes	Block inputs: zero padding in Goal, live in Push	Partly

- The bound is an input because nothing else changes on the threshold axis (Saute's augmentation, without its absorbing state)
- With two bounds per sequence, the bound acts as a task label, not a learned bound → behaviour map

Normalising the budget channel

Normalisation	Reward	Cost
by the bound	13.76	56.24
by a constant (25), used	47.74	13.35

Loosening to $d = 25$, strength 10000, original level, one seed (Table A.11).

A4 · What F_i is: importance measures

$$F_i = \frac{1}{N} \sum_{n=1}^N w_n \left(\frac{\partial}{\partial \theta_i} \log \pi_{\theta^*}(a_n | s_n) \right)^2$$

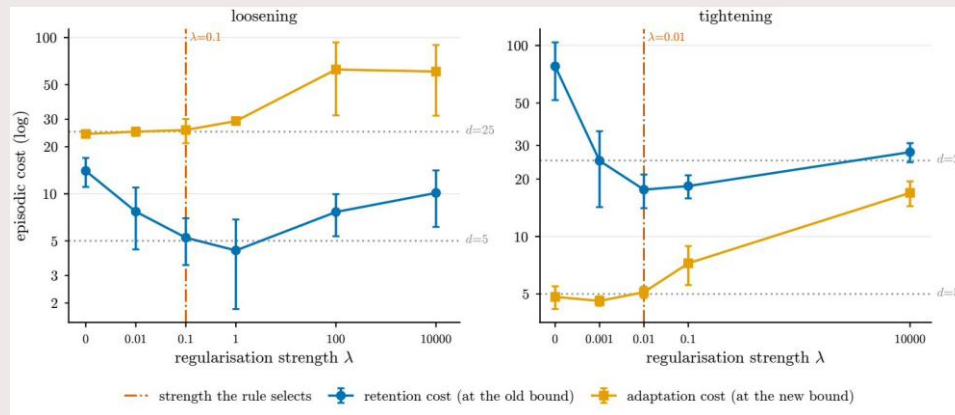
- **Standard Fisher:** squared gradient of the anchor policy's action log-probability; how sensitive its action choices are to each weight. Computed, not tuned
- **Cost-aware arm:** each sample weighted by the squared cost advantage $G_c - V_c(s)$, from the cost critic
- **Uniform:** $F_i = 1$
- **CF-EWC** (Coursey et al.): weight $1/(1 + c)$. Compared as a map only, never run as an arm

Map vs standard Fisher	Mean cos	Min cos	Top-10%
CF-EWC (cost-weighted)	1.0000	1.0000	0.998
Reward-advantage	0.9985	0.9670	0.936
Cost-aware arm (state-dep.)	0.9899	0.9040	0.832
Cost-advantage	0.9847	0.8570	0.791

- Cosine similarity: 1 = same relative stiffness on every weight (direction of the 5,572-long vector). 1.0000 is rounded to four decimals
- Zero-cost samples: 94.2–99.1% of a safe agent's data, so $1/(1 + c) = 1$ almost everywhere
- The cost-aware arm's map does differ (min 0.90); its retention differs from Fisher by at most 1.2 standard errors

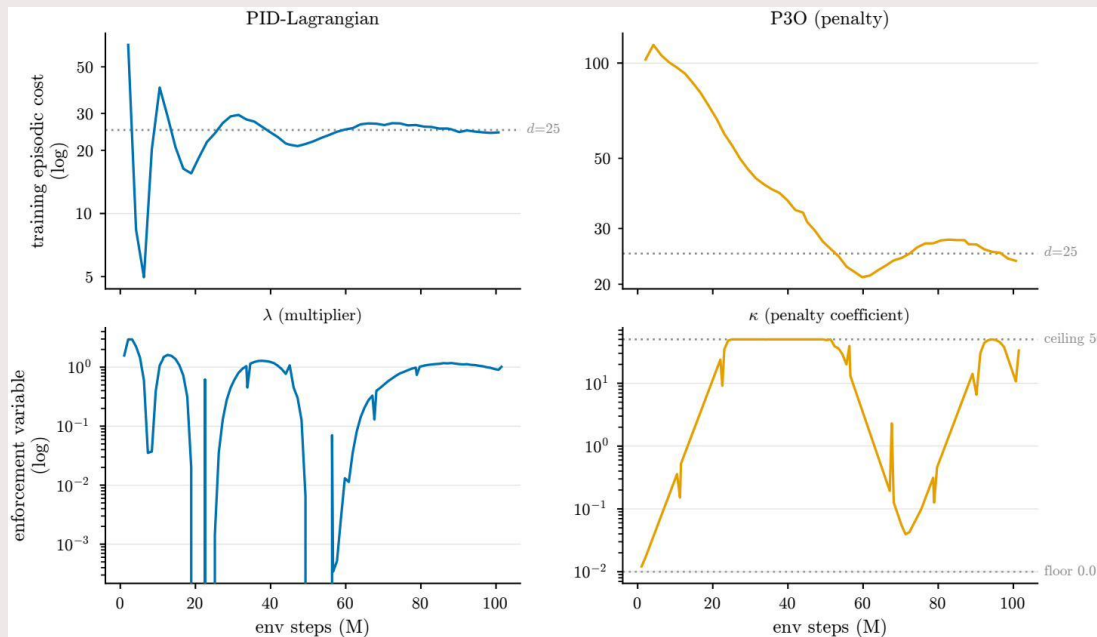
A5 · Choosing the EWC strength

- **1.** Keep strengths with new-bound cost $\leq 1.05 \times$ bound
- **2.** Of those, keep the ones meeting the old bound
- **3.** Of those, take the highest reward
- If step 2 is empty, take the lowest old-bound cost: loosening and L11 $\rightarrow$ L14
- L11 $\rightarrow$ L14: $\lambda = 10$ by judgement; strengths 1 to 10000 lie within 3 cost units
- Swept at 2–3 seeds, then extended to 5: reported values share seeds with the selection



Loosening: the strength with the lowest old-bound cost breaches the new bound. Tightening: it does not (Fig 6.4).

A6 · Why P3O



- Five configurations, each trained once with PID-Lagrangian and once with P3O: final reward and cost near-identical in 4 of 5
- Neither family visibly more stable (single seeds)
- Chosen on mechanism: a Lagrange multiplier integrates the size of past violations; κ moves by a fixed ratio on the sign of the violation and is clipped, so it cannot wind up within a phase

A7 · Other methods considered

Pruned	Reward	Cost	Cost ÷ bound
none	30.26	4.85	0.97
50%	16.78	15.55	3.11
70%	1.18	268.77	53.8

Magnitude pruning of a trained anchor ($d = 5$), one run, no retraining (Table 5.3). PackNet retrains before freezing, so this suggests little spare capacity; it does not rule PackNet out.

- PackNet on a larger network is the named follow-up; it also tests the mechanism, since freezing weights should show no direction asymmetry
- A-GEM forgets far more than regularisers on MEAL; replay memory grows with the data kept

Training on all bounds at once?

- CCPO samples bounds throughout training: phase 1 would already train on phase 2's bound, so neither adaptation nor retention could be measured
- The deployment setting assumes the new bound is not known when the agent is certified
- Being told the bound is not enough: plain retraining breaks the old bound while reading it
- Multi-bound training is a baseline worth adding in future work

B1 · How the levels were built

Level	Hazards	Coverage	Goal speed	Reward gap	Runs
L4	20 × 0.65	106%	0.00	+0.01	1
L5	12 × 0.50	37.7%	0.00	+7.68	1
L6	40 × 0.18	16.3%	0.00	-1.03	1
L7	40 × 0.18	16.3%	0.30	+5.65	1
L12	40 × 0.18	16.3%	0.40	+5.39	1
L10	40 × 0.18	16.3%	0.50	+5.82	1
L13	40 × 0.18	16.3%	0.65	+7.44	1
L9	40 × 0.18	16.3%	0.80	-0.75	1
L8	55 × 0.16	17.7%	0.50	+11.73	3
L15	55 × 0.16	17.7%	0.65	+10.89	3
L11	40 × 0.22	24.3%	0.50	+14.44	3
L14	70 × 0.16	22.5%	0.50	+27.37	5

- Reward gap = reward at $d = 25$ minus $d = 5$, paired per seed
- A still goal (L6) or a goal faster than the robot (L9) removes the gap
- Many small hazards beat fewer large ones: L14 vs L11
- Original CRAX levels: little or no gap (Push L2: -9.29)

B2 · Validated anchors

L14 at four bounds (5 runs)

Bound	Reward	Cost	Cost ÷ d
5	31.73 ± 2.06	4.30 ± 0.59	0.86
10	48.83 ± 2.93	9.70 ± 0.70	0.97
15	55.22 ± 5.09	13.36 ± 1.09	0.89
25	59.10 ± 5.71	24.70 ± 2.68	0.99

Axis endpoints at d = 25 (5 runs)

Level	Reward	Cost	Cost ÷ d
L1 (original)	80.00 ± 5.77	24.43 ± 1.74	0.98
L8	61.38 ± 6.75	24.61 ± 1.28	0.98
L11	57.99 ± 7.05	24.02 ± 1.87	0.96
L14	59.10 ± 5.71	24.70 ± 2.68	0.99
L16	56.32 ± 5.74	24.15 ± 1.76	0.97
Goal L1 (padded)	35.44 ± 0.24	23.92 ± 1.53	0.96

Every anchor used satisfies its bound. Two unused combinations sit just above: L15 at d = 25 (1.07) and L11 at d = 15 (1.03).

B3 · Configuration

Parameter	Value
Optimiser	Adam
Learning rate	1×10^{-3} *
Entropy coefficient	1×10^{-2} *
Discount γ	0.99
GAE λ	0.95
PPO clipping	0.3
Reward scaling	0.1
Unroll length	8
Batch size	1024

Parameter	Value
Minibatches / updates	32 / 6 per batch
Parallel environments	2048
Steps per phase	5×10^8
Policy	MLP 4×32 (5,572 weights)
Critics	MLP 5×256 , reward and cost
Penalty κ	0.01 start, clipped [0.01, 50]
Episode length	2000 Push, 1000 Goal
Evaluation	deterministic, 128 episodes
Seeds	5 unless stated

Asterisk: differs from the published CRAX defaults (see B4). A 500M-step run takes three to four hours on one GPU.

B4 • CRAX defaults

Setting (original Push level, d = 25)	Reward	Runs
Published defaults	29.03 ± 14.12	5
Final: learning rate 1e-3 + entropy 1e-2	74.40	1
Learning rate only	65.78	3
Entropy only	65.29	1
Discount 0.995–0.999	5.92–11.13	1
Longer unroll	6.79	1
More updates per batch	21.73	1
Further learning-rate increase	0.10	1

- Only two values changed; everything else as published
- Final configuration, 5 runs: 80.00 ± 5.77, against the paper's 41.0 for P3O
- Two 1B-step runs reach 69.81 and 72.53, still climbing, against 500M means of 29.03 and 43.16
- The anchors used here may not have fully converged

C1 · Threshold axis, all arms

Arm	R old	C old	p_{ret}	R new	C new	p_{adp}
Loosening 5 → 25: anchor	31.73 ± 2.06	4.30 ± 0.59	0.86	—	—	—
plain retraining	45.02 ± 6.09	14.02 ± 2.95	2.80	61.91 ± 7.43	24.16 ± 0.93	0.97
standard Fisher	27.04 ± 4.58	5.24 ± 1.74	1.05	34.13 ± 9.98	25.61 ± 4.53	1.02
state-dep. baseline	22.94 ± 8.88	6.28 ± 2.37	1.26	27.62 ± 13.96	26.79 ± 3.15	1.07
uniform anchoring	27.42 ± 4.29	5.20 ± 2.00	1.04	35.11 ± 5.37	27.99 ± 2.90	1.12
Tightening 25 → 5: anchor	59.10 ± 5.71	24.70 ± 2.68	0.99	—	—	—
plain retraining	44.29 ± 10.82	77.79 ± 26.03	3.11	35.27 ± 3.22	4.83 ± 0.66	0.97
standard Fisher	46.98 ± 5.79	17.59 ± 3.52	0.70	33.31 ± 3.61	5.11 ± 0.35	1.02
state-dep. baseline	46.29 ± 5.38	16.38 ± 3.43	0.66	32.42 ± 4.16	5.51 ± 0.79	1.10
uniform anchoring	49.37 ± 4.65	15.56 ± 1.76	0.62	38.96 ± 1.78	7.33 ± 0.76	1.47

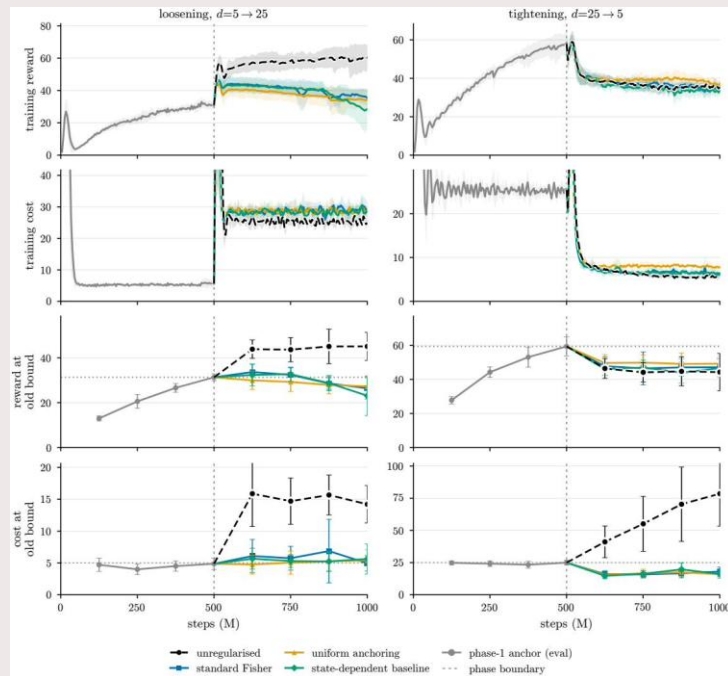
L14, 5 runs. $\lambda = 0.1$ loosening, 0.01 tightening. R, C = episodic reward and cost at the old and new bound (Table 6.1).

C2 · Threshold strength sweep

λ	C old	R new	C new	Runs
Loosening 0	14.02	61.91	24.16	5
0.01	7.69	48.70	25.00	3
0.1 (selected)	5.24	34.13	25.61	5
1	4.34	20.75	29.21	3
100	7.65	15.42	62.48	3
10000	10.13	16.97	60.64	5
Tightening 0	77.79	35.27	4.83	5
0.001	24.92	32.81	4.60	3
0.01 (selected)	17.59	33.31	5.11	5
0.1	18.37	33.36	7.24	3
1 / 10 / 100	20.99 / 25.95 / 27.51	40.93–44.50	11.45–18.13	2
10000	27.65	43.09	16.93	5

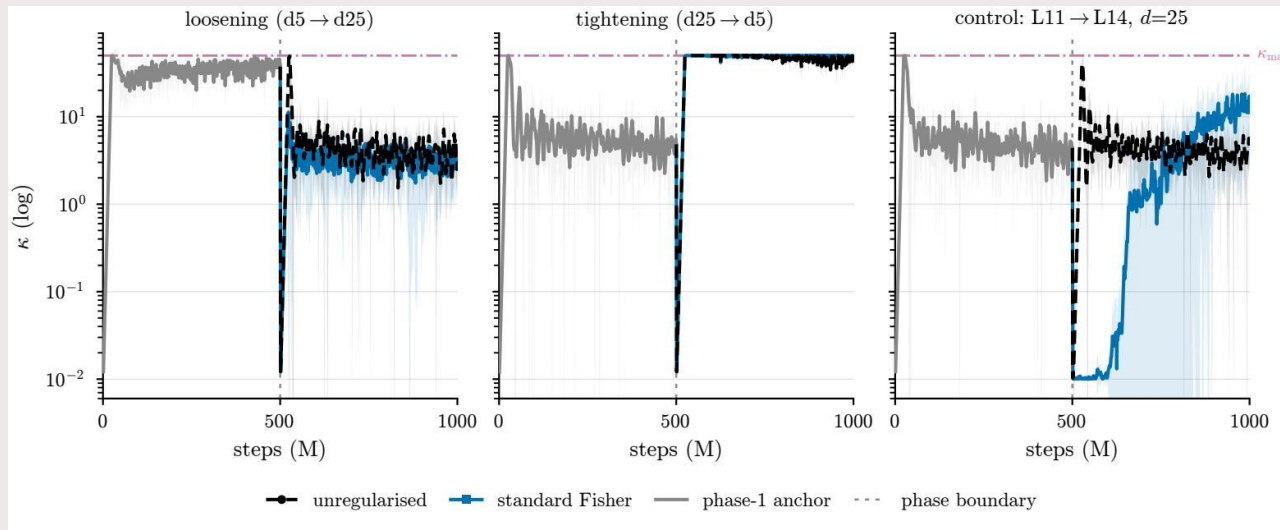
- Loosening: 0.32 cost units of retention per reward point given up
- Choosing 0.01 instead: 0.48, still giving up 21% of reward
- Best retention (4.34) breaches the new bound by 16.8%
- At 10000: loosening forfeits 73% of reward and breaches the new bound; tightening fails to tighten

C3 · Retention through training



- Loosening: plain retraining is about three times over the old bound at the first checkpoint and stays there
- Tightening: it degrades steadily and is still rising when the phase ends
- Every EWC arm is flat in both directions
- Stopping early would not have preserved the old certification when loosening

C4 · Safety pressure over training



- Median κ over 163 phase-2 runs: 2.05 loosening, 50.00 tightening; every tightening run reaches the ceiling
- Plain and EWC arms follow near-identical κ within each direction: the two mechanisms act separately

C5 · Difficulty, per transition

Transition	Predicted cost ÷ bound	Retention (old level)		Adaptation (new level)		Old bound
		reward, % of anchor	cost ÷ bound	reward, % of anchor	cost ÷ bound	
L14 → L1	9.46	61% → 82%	8.61 → 0.82	60% → 79%	0.95 → 0.79	Restored
L11 → L14	2.20	112% → 97%	2.28 → 1.31	107% → 86%	0.92 → 0.95	Still above
L14 → L8	1.50	113% → 94%	1.51 → 0.97	110% → 92%	0.98 → 0.65	Restored
L14 → L16	1.22	107% → 94%	1.07 → 0.95	114% → 98%	1.03 → 0.92	Restored
L8 → L11	0.41	93% → 82%	0.42 → 0.39	103% → 92%	0.98 → 1.00	Never broken

Each cell: plain retraining → EWC. Reward as % of the anchor trained on that level; cost ÷ bound (25), green ≤ 1. Predicted before any retraining. L14 → L16 and L8 → L11 at 3 runs.

SQ2: plain retraining loses up to 8.61 × the bound, as predicted; EWC restores three of four breaches.

C6 · Difficulty, all arms

Transition	Arm	λ	C old (ρ_{ret})	R new	C new	Runs
L14 $\rightarrow$ L1	plain	—	215.13 (8.61)	48.30 $\pm$ 44.18	23.76	5
	Fisher / uniform / state-dep.	1	20.46 / 19.09 / 19.08	63.02 / 62.32 / 59.29	18.6–19.9	5
L11 $\rightarrow$ L14	plain	—	56.89 (2.28)	63.03 $\pm$ 7.36	23.00	5
	Fisher / uniform / state-dep.	10	32.87 / 34.25 / 35.25	50.59 / 50.39 / 49.71	23.4–24.8	5
L14 $\rightarrow$ L8	plain	—	37.76 (1.51)	67.64 $\pm$ 6.57	24.44	5
	Fisher / uniform / state-dep.	10	24.24 / 25.27 / 24.25	56.72 / 56.67 / 56.82	16.3–16.9	5
L14 $\rightarrow$ L16	plain	—	26.75 (1.07)	64.23 $\pm$ 8.04	25.65	3
	Fisher	100	23.72 (0.95)	55.43 $\pm$ 5.56	23.11	3
L8 $\rightarrow$ L11	plain	—	10.59 (0.42)	59.69 $\pm$ 7.83	24.47	3
	Fisher	0.1	9.75 (0.39)	53.21 $\pm$ 8.71	25.09	3

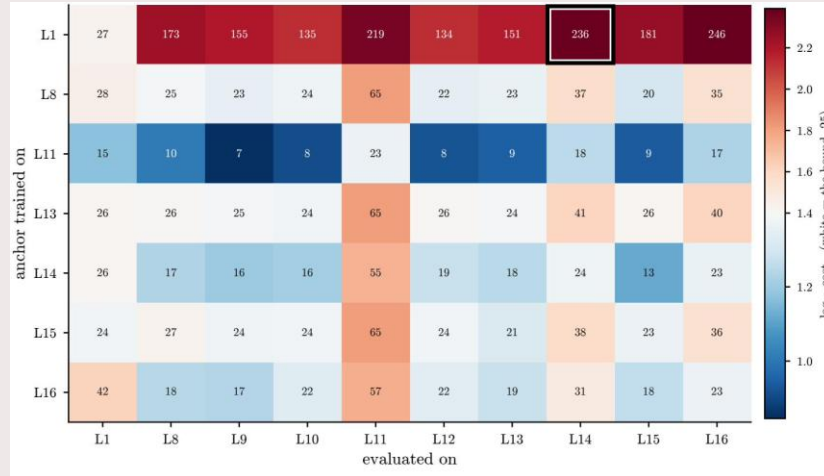
Bound 25 throughout (Table 6.3). L14 $\rightarrow$ L1 is the only pair where EWC also raises adaptation reward.

C7 · L11 → L14: no λ clears it

Arm	λ	C old	R new	C new	Runs
plain	—	56.89 ± 8.24	63.03	23.00	5
Fisher	0.01	44.39 ± 3.82	58.57	23.77	3
Fisher	0.1	31.38 ± 9.79	43.21	21.17	3
Fisher	1	35.37 ± 2.41	51.49	23.20	3
Fisher	10	32.87 ± 4.67	50.59	23.75	5
Fisher	100	32.52 ± 2.97	50.07	24.10	3
Fisher	1000	34.18 ± 2.56	50.55	25.19	3
Fisher	10000	34.58 ± 5.77	50.59	24.68	3

- Bound 25: every strength lowers the violation, none clears it
- $\lambda = 0.1$ has the lowest mean, pulled by one run whose reward collapsed
- A possible cause: L11's larger hazards (radius 0.22) may need wider berths than small-hazard policies learned

C8 · Predicting forgetting



Anchor cost at $d = 25$, one evaluation per cell (Table A.6).

Transition	Predicted	Observed	Error	Runs
L14 → L1	236.5	215.13	-9%	5
L11 → L14	55.0	56.89	+3%	5
L14 → L8	37.4	37.76	+1%	5
L14 → L16	30.6	26.75	-13%	3
L8 → L11	10.2	10.59	+4%	3
L1 → L14	26.0	27.66	+6%	1
L16 → L14	22.9	22.68	-1%	1

- Predicts plain retraining only
- 5 of 7 pairs were chosen from the matrix; the two single-run pairs test it on unselected transitions

C9 · Task axis, all strengths

Direction	Arm	R old	C old (ρ_{ret})	R new	C new	Runs
Goal → Push	anchor (Goal)	35.44	23.92 (0.96)	—	—	5
	plain	1.12	146.53 (5.86)	69.42	24.23	3
	$\lambda = 0.001 / 0.01 / 0.1$	3.94 / 3.78 / 15.62	117.14 / 49.93 / 60.57	58.43 / 26.17 / 5.48	17.0–21.9	3
	$\lambda = 1 / 100$	28.99 / 18.47	62.39 / 45.59	0.14 / 1.42	0.32 / 15.60	1
Push → Goal	anchor (Push L1)	80.00	24.43 (0.98)	—	—	5
	plain	−0.22	306.85 (12.27)	34.52	25.40	3
	$\lambda = 0.001 / 0.01 / 0.1$	7.59 / 21.02 / 27.16	88.39 / 99.56 / 104.03	34.36 / 32.25 / 28.13	22.8–26.0	3
	$\lambda = 1 / 100$	20.86 / 17.67	118.00 / 123.23	16.15 / 0.11	20.49 / 18.40	1

Bound 25, Goal L1 $\leftrightarrow$ Push L1, observations aligned by zero-padding (Table 6.4). Plain retraining per-run costs: 40–334 and 83–671, so the claim holds in direction and order of magnitude only.

C10 · Chains: difficulty, control

Arm	L11 (R / C)	L14 (R / C)	L1 (R / C)	Runs
no phase-3 penalty	28.54 / 211.48	27.60 / 208.91	33.11 / 42.50	5
standard Fisher	49.31 / 30.68	39.66 / 22.50	61.30 / 12.63	5
uniform anchoring	49.68 / 31.99	40.64 / 23.34	61.76 / 14.10	5
state-dep. baseline	45.00 / 47.23	36.82 / 39.04	48.52 / 15.42	5

L11 → L14 → L1 at bound 25; $\lambda_1 = 10$, $\lambda_2 = 1$. Cost on L11 at the start → end of phase 3: no penalty 163.44 → 210.64; Fisher 36.57 → 30.22; uniform 37.40 → 29.25 (backward transfer on cost).

Control chain L14 → L1 → L16	L14 (R / C)	L1 (R / C)	L16 (R / C)	Runs
no phase-3 penalty	63.17 / 25.60	66.89 / 30.47	64.28 / 24.07	3
standard Fisher, $\lambda_2 = 1$	53.12 / 26.07	65.35 / 24.07	52.35 / 25.17	3

The control chain has little to forget, so it is excluded from the claim (Table 6.5).

C11 · Chains: threshold

Arm	d = 25 (R / C)	d = 5 (R / C)	Runs
no phase-3 penalty	53.36 / 22.64	32.01 / 13.58	5
standard Fisher	48.51 / 21.17	37.46 / 5.79	5
uniform anchoring	48.87 / 25.01	38.13 / 6.64	5
state-dep. baseline	49.47 / 22.94	37.18 / 6.64	5

Return chain $d = 25 \rightarrow 5 \rightarrow 25$ on L14; $\lambda_1 = 0.01$, $\lambda_2 = 1$. $d = 25$ is both retention and adaptation here, so only two independent points.

Arm	d = 15 (R / C)	d = 25 (R / C)	d = 10 (R / C)	Runs
no phase-3 penalty	59.73 / 17.95	61.41 / 75.90	57.49 / 9.99	5
standard Fisher	51.62 / 11.36	54.27 / 21.02	48.01 / 9.63	5
uniform anchoring	50.97 / 12.68	54.57 / 22.37	47.12 / 10.25	5
state-dep. baseline	51.93 / 12.58	54.88 / 21.51	48.16 / 9.96	5

Three-bound chain $d = 15 \rightarrow 25 \rightarrow 10$ on L14; $\lambda_1 = 0.1$, $\lambda_2 = 1$ (Table 6.5).

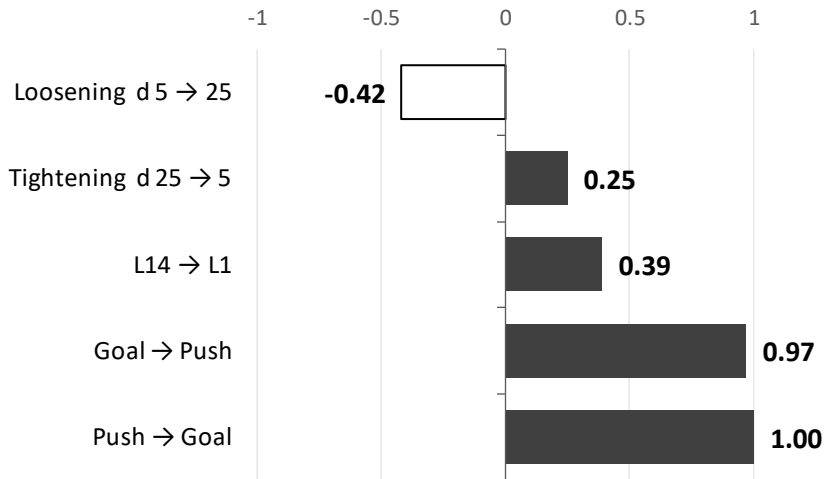
C12 · Cost during retraining

Group	Runs	Satisfy $\rho_{c,T} \leq d$
Phase-1 anchors	229	0
Threshold axis, phase 2	163	23
Difficulty axis, phase 2	160	59
Task axis, phase 2	28	0
Three-phase chains	99	53

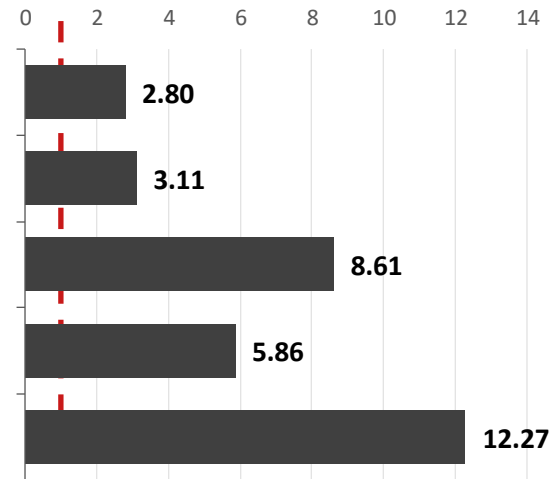
- Cost accumulated over training, per Ray et al. (2019)
- Loosening: EWC arms 29.41–29.84 per episode against plain retraining's 28.24; tightening: 8.15–9.26, indistinguishable
- No threshold arm satisfies its bound during training: $1.1\text{--}1.9 \times$ the bound
- EWC protects the final policy, not the learning process. Anchors score 0 by construction: they start untrained

C13 · Reward rises as the bound breaks

Reward-based forgetting (0 = kept, 1 = lost)



ρ_{ret} (1 = bound)



Plain retraining, same runs scored two ways. Negative forgetting: more reward on the old setting than the anchor (31.73 → 45.02). Table 7.1.

The standard forgetting metric reports an improvement where the certificate breaks.

V1 · Seeds and statistics

- Values are means $\pm$ standard deviation over runs: 5, or 3 on the task axis, L14 $\rightarrow$ L16 and L8 $\rightarrow$ L11
- At 5 runs, the standard error of a difference between two arms is about 0.63 of one arm's standard deviation
- No hypothesis tests: at 3 to 5 runs they would have little power
- Between EWC versions, retention gaps are at most 1.2 standard errors; plain vs EWC: 4.6 to 5.7 standard errors when the bound changes
- Strengths were selected at 2–3 runs and extended to 5, so reported values share runs with the selection
- Runs from different hyperparameter settings are never pooled
- Reading of cost $\div$ bound on the slides: ≤ 1 holds; 1.00–1.05 at the bound, the tolerance the λ rule allows on the new bound; above 1.05 broken. Loosening EWC: 5.24 ± 1.74 against 5

V2 · Forgetting or conditioning?

Run (original level, bound 5)	Cost at d = 5	Drift
Anchor	3.93 ± 0.59	—
+ one phase at the same bound	5.19 ± 0.79	+1.26
+ one phase at d = 25	19.89 ± 4.11	+15.96

Same-bound control: about nine tenths of the drift comes from the bound change. Original level, default hyperparameters, so indicative for L14.

- Anchors meet their own bound (0.86–0.99) before phase 2
- The breach builds up during retraining (appendix C3)
- EWC, which pulls weights back to the anchor, restores it after tightening

What remains open

- Phase 2 trains only at the new bound, so the network's response to the old bound's input is no longer trained
- The budget channel is divided by 25: a policy trained at d = 5 sees it only in the bottom fifth, so its behaviour at d = 25 is extrapolation
- Pure conditioning is ruled out; a mix of forgetting and input extrapolation is not (Section 7.7)

V3 · Does seeing the change help?

Axis	Change visible to the agent?	Best retention ratio reached
Threshold	Yes	0.62 to 1.04
Difficulty	No	0.39 to 1.31 across five pairs
Task	Partly	2.00 and 3.54

- Task retains worst; threshold and difficulty do not differ measurably, so visibility does not order the axes
- The axes also differ in the size of the change, and the task axis changes episode length and reward scale

V4 · Coursey et al. compared

	Coursey et al. (2026)	This thesis
Safe RL in the combination	Proposed methods on PPO, no bound enforced	P3O, a constrained method, with EWC
What changes	Dynamics; bound fixed	The bound itself, dynamics, task
Agents before the change	Total cost in the thousands for PPO, PPO+EWC, CF-EWC	Certified: 0.86–0.99 of the bound
Safety measured as	Total cost during training	Test cost against the old bound (p_{ret})
CF-EWC	Performs like PPO+EWC; no explanation	Identical to the Fisher on safe agents (cosine 1.0000)

Coursey et al. (2026), Tables 1–3: Damaged HalfCheetah, Damaged Ant, Safe Continual World.

They also propose Safe EWC (cost folded into reward); it was not among the arms tested here.

V5 · What Coursey et al. actually did

What changes	the environment; the bound stays fixed
Their method	PPO + EWC — a fixed cost penalty, no constrained method
Were the agents safe?	no — cost in the thousands, far above their bounds
Safety measured as	total cost during training, not vs a bound
Their two arms	EWC and CF-EWC; both matched plain PPO+EWC
Our answer	on a safe agent $w = 1$, so CF-EWC equals EWC (cosine 1.0000)

Coursey et al. never had a bound-satisfying agent to preserve — on one, CF-EWC collapses to plain EWC.

V6 · Deployment and regulation

- A certification applies to the policy as certified (Hawkins et al., 2021); here it did not survive plain retraining on most changes, nor EWC when loosening, and nothing measured at the current bound showed it
- Check: after every round of training, test against every bound ever certified; 128 episodes per bound against 500M training steps; needs a record of superseded bounds
- EU AI Act Art. 43(4): changes declared in advance are not "substantial"; the FDA's predetermined change control plan works the same way
- Re-certification should follow any change of specification, not only elapsed time or an observed violation (Denney et al., 2015)
- A change in the environment gives no event that would prompt the check

V7 · Future work

- **Parameter isolation on a larger network:** tests whether the asymmetry belongs to gradient-based retention or to the constraint change itself
- **Size of the bound change:** anchors at $d = 10$ and 15 already exist; only the adaptation phases are missing
- **Separate the explanations:** a higher κ ceiling or a violation-proportional controller; a conditioned vs an unconditioned policy
- **A cost-aware importance measure** built on a statistic accumulated over the episode
- **Longer sequences** (Online EWC) and **composite changes**; a Lagrangian safety method
- **Beyond safety:** a fairness or latency budget revised between rounds raises the same question

V8 · Plan versus final thesis

Planned	Final	Reason
Saute RL as the safety method	P3O with Saute's budget input only	Saute's absorbing state collapses reward on long episodes
EWC, CLEAR, PackNet	EWC family only	Little capacity to partition; replay memory grows
A composite metric	Separate cost-side metrics	A combined score hides a safety breach
Threshold and task changes	Plus difficulty changes, on custom levels	Original levels have little reward–cost trade-off